

Find the Error: Rounding and Place Value

Answer Key

Grades 4–5 Mathematics

Quick Answers

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| 1. 3000 | 6. the correct number = 50307, — |
| 2. 40000 | 7. 748 rounds to = 700, 651 rounds to = 700, 762 |
| 3. the correct answer = 5000, — | rounds to = 800, 685 rounds to = 700, — |
| 4. the correct product = 1200, — | 8. 452 rounded = 450, 448 rounded = 450, — |
| 5. 10000 | |
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What is verified

3 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 3, 4, 6, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Grades 4–5 Mathematics

4.NBT.A.1 – *problem 4*

4.NBT.A.2 – *problem 6*

4.NBT.A.3 – *problems 1, 2, 3, 5, 7, 8*

Difficulty 1–4 of 5 across 17 tagged checks.

Common wrong answers

1. *If they got 2900: rounded the 9 hundreds down instead of carrying 29 hundreds up to 30*
 2. *If they got 39000: rounded down to 39,000, ignoring that the hundreds digit 7 rounds 39 thousands up to 40*
 3. *If they got 4900: kept the 49 hundreds and rounded down, ignoring that the tens digit 7 rounds 49 hundreds up to 50*
 4. *If they got 120: multiplied 3 by 4 and attached only one zero instead of two*
 5. *If they got 9900: rounded the 99 hundreds down to 9,900 instead of carrying up to 100 hundreds*
 6. *If they got 5307: pushed the nonzero digits together and dropped the placeholder zeros for the empty thousands and tens places*
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- 1.** Round 2,963 to the nearest hundred.

Solution: The hundreds digit is 9; the tens digit is $6 \geq 5$, so round up. 29 hundreds become 30 hundreds — the carry crosses into the thousands: $2,963 \rightarrow 3,000$. 3000

2. Round 39,725 to the nearest thousand.

Solution: The thousands digit is 9; the hundreds digit is $7 \geq 5$, so round up. 39 thousands become 40 thousands: $39,725 \rightarrow 40,000$. 40000

3. Leo rounds 4,972 to the nearest hundred and writes 4,900.

Solution: The tens digit of 4,972 is 7, which says round the hundreds *up*. Rounding 49 hundreds up gives 50 hundreds, so the correct answer is 5,000. 5000

Full-credit explanation: Leo rounded down (kept 49 hundreds) even though the tens digit 7 demands rounding up, and he missed that the carry crosses the thousands boundary.

4. Mia multiplies 30×40 and writes 120.

Solution: $30 \times 40 = (3 \times 10) \times (4 \times 10) = 12 \times 100 = 1,200$. 1200

Full-credit explanation: there are *two* factors of ten — one from 30 and one from 40 — so the basic fact $3 \times 4 = 12$ needs *two* placeholder zeros. Mia attached only one.

5. Round 9,984 to the nearest hundred.

Solution: The tens digit is $8 \geq 5$, so the 99 hundreds round up to 100 hundreds: $9,984 \rightarrow 10,000$. Every digit changes — that is what rounding across a 9 does. 10000

6. Jon writes $50,000 + 300 + 7 = 5,307$.

Solution: Fill the places: 5 ten-thousands, 0 thousands, 3 hundreds, 0 tens, 7 ones. The empty thousands and tens places need placeholder zeros: $50,000 + 300 + 7 = 50,307$. 50307

Full-credit explanation: Jon pushed the nonzero digits together and dropped the zeros, which moved the 5 from the ten-thousands place to the thousands place.

7. Circle all that round to 700 (nearest hundred): 748, 651, 762, 685.

Solution: Round each one. 748: tens digit $4 < 5$, rounds down to 700

. 651: tens digit $5 \geq 5$... careful, 651 is *above* 650, between 600 and 700 and past the midpoint, so it rounds up to 700

. 762: tens digit $6 \geq 5$, rounds up to 800

. 685: tens digit $8 \geq 5$, rounds up to 700

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Circle 748, 651, and 685. Do *not* circle 762 — it rounds to 800.

8. Ava: “452 and 448 round to the same ten, so they are equal.”

Solution: 452: ones digit $2 < 5$, rounds down to 450

. 448: ones digit $8 \geq 5$, rounds up to 450

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Full-credit explanation: rounding replaces each number with the *nearest* ten, so two different numbers can land on the same rounded value. Sharing a rounded value does not make them equal: $452 \neq 448$ — they just both live closest to 450.